

CHE 341: Chemical Process Control

Lecture #2

Chemical Process Control and Process Safety

Dr. Michael Caracotsios

**Clinical Associate Professor
Chemical Engineering Department**

Process Control Objectives

- Assure Safe Operations
- Maintain Quality of the desired Products (on spec)
- Achieve significant reduction in the cost of production

Although all three goals are very important assuring safe operations is by far the most critical since it involves human lives and possible devastation effects on the ecosystem and surrounding communities

Example

BP Texas City Refinery Explosion
(Failure of Level Control). Taken
from the Chemical Safety and
Hazard Investigation Board

<https://www.csb.gov/bp-america-refinery-explosion/>

Discussion of Recommendations

- From [API] Revise API Recommended Practice 521, **Guide for Pressure Relieving and Depressurizing Systems** to ensure that the guidelines: - Identifies overfilling vessels as a potential hazard for evaluation in selecting and designing pressure relief and disposal systems; - Addresses the need to adequately size disposal drums for credible worse-case liquid relief scenarios, based on accurate relief valve and disposal collection piping studies; - Warns against the use of atmospheric blowdown drums and stacks attached to collection piping systems that receive flammable discharges from multiple relief valves and urges the use of appropriate inherently safer alternatives such as a flare system
- From [BP] Evaluate your refinery process units to ensure that critical process equipment is safely designed. At a minimum, a. Ensure that distillation towers have **effective instrumentation and control systems to prevent overfilling such as multiple level indicators and appropriate automatic controls.** b. Configure control board displays to clearly indicate material balance for distillation towers.
- **Classroom Discussion**